

KAGGLE HACX '26 — DATA SPRINT TO THE PEAK

Project Title: Eye Disease Classification

1. Problem Statement

Eye diseases such as Cataract, Diabetic Retinopathy, and Glaucoma are leading causes of vision loss worldwide. Early and accurate detection is critical for preventing permanent blindness. Manual diagnosis requires specialist expertise and is time-consuming. This project builds an automated deep learning system to classify retinal fundus images into four categories:

- Cataract
- Diabetic Retinopathy
- Glaucoma
- Normal (Healthy)

2. Dataset

- **Dataset Name:** Eye Diseases Classification
- **Source:** KaggleHacX '26 Competition Data
- **Total Images:** 4,217
- **Train Set:** 3,376 images
- **Validation Set:** 841 images
- **Classes:** 4 (Cataract, Diabetic Retinopathy, Glaucoma, Normal)
- **Class Distribution:** Approximately balanced (~1,000 images per class)
- **Image Format:** JPG/PNG, varying resolutions

3. Preprocessing Steps

All images were preprocessed before feeding into the model:

- Resized to 224×224 pixels
- Pixel values normalized to [0, 1] by dividing by 255
- 80/20 train-validation split applied
- Data augmentation applied on training set:
 - a. Horizontal flip
 - b. Rotation up to 15°
 - c. Zoom up to 10%

4. Model Choice

We used Transfer Learning with EfficientNetB0 pretrained on ImageNet.

Why EfficientNetB0?

- State-of-the-art accuracy with minimal parameters
- Pre-trained on 1.2M ImageNet images — already knows edges, textures, shapes
- Ideal for medical image classification
- Fast training — critical for hackathon time constraints

Custom Classification Head Added:

- GlobalAveragePooling2D
- Dropout (0.3)
- Dense layer — 128 neurons, ReLU activation
- Dropout (0.2)
- Output Dense layer — 4 neurons, Softmax activation

5. Training Strategy

Phase 1 — Warm-up (Frozen base):

- Base model layers frozen
- Only custom head trained
- Optimizer: Adam (lr=0.001)
- Epochs: 15 (Early stopping triggered at Epoch 4)

Phase 2 — Fine-tuning (Unfrozen base):

- All layers unfrozen
- Lower learning rate: 1e-4
- Optimizer: Adam
- Epochs: 20 (Best weights from Epoch 17 restored)
- Loss Function: Categorical Crossentropy
- Batch Size: 32
- Hardware: Kaggle GPU Tesla T4

6. Callbacks Used

- EarlyStopping(patience=5) — prevents overfitting, restores best weights
- ReduceLROnPlateau(patience=3, factor=0.3) — reduces learning rate when val_loss plateaus.

7. Validation Approach

- 20% holdout validation set (841 images)
- Monitored val_accuracy and val_loss each epoch
- Final model selected based on best validation weights (Epoch 17)
- Evaluated using classification report and confusion matrix

8. Training Metrics

EPOCH	TRAIN ACCURACY	VALUE ACCURACY
1(Phase 2)	57.69%	26.16%
4	92.17%	44.47%
7	94.44%	84.66%
14	95.19%	97.58%
17(BEST)	95.35%	84.66%

9. Final Results

CLASS	PRECISION	RECALL	F-1 SCORE
Cataract	95%	94%	94%
Diabetic Retinopathy	99%	99%	99%
Glaucoma	94%	80%	87%
Normal	83%	95%	89%
Overall Accuracy	92%		

10. Demo Application

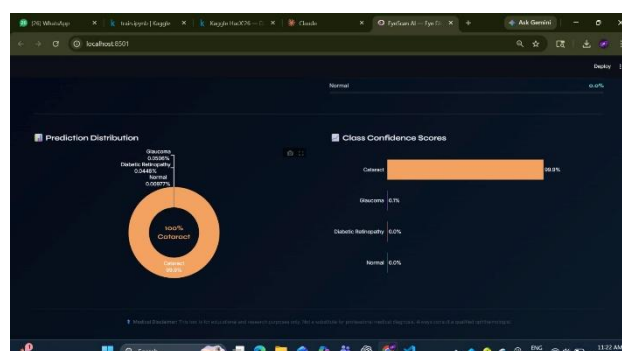
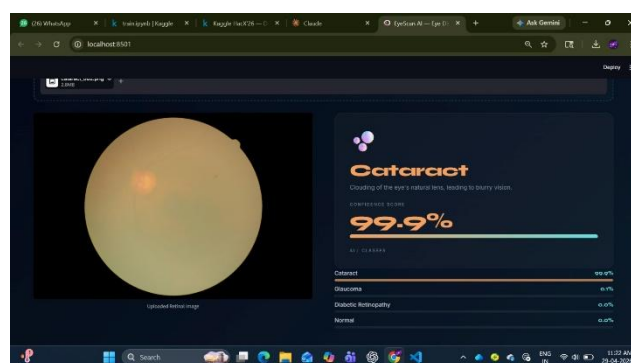
Built using Streamlit framework.

Features:

- Upload any retinal fundus image (JPG/PNG)
- Instant classification with confidence score
- Visual confidence bars for all 4 classes
- Clean, professional UI

Tech Stack:

- Python 3.x
- TensorFlow/Keras
- Streamlit
- NumPy, PIL



11. Conclusion

Our EfficientNetB0-based transfer learning model achieved **92% overall accuracy on eye disease classification. The model performs exceptionally well on Diabetic Retinopathy (99% F1) and Cataract (94% F1). The Streamlit demo app provides an intuitive interface for real-world usage.

Team Name: DATADUO

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